

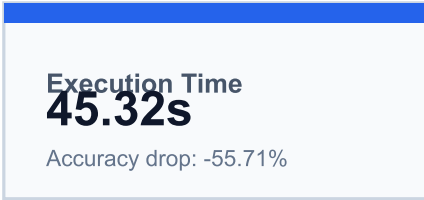
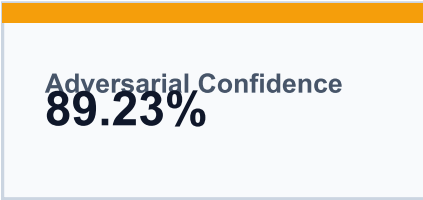
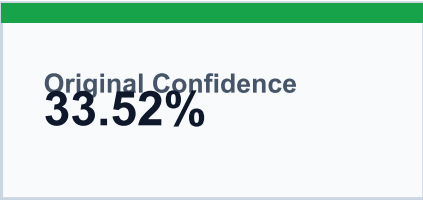
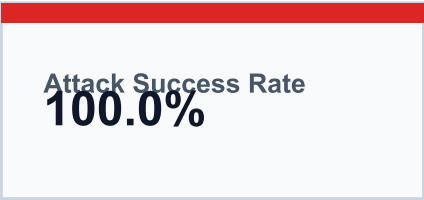
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-18

Attack Type: PGD

Images Evaluated: 10



Threat Level Assessment

HIGH RISK

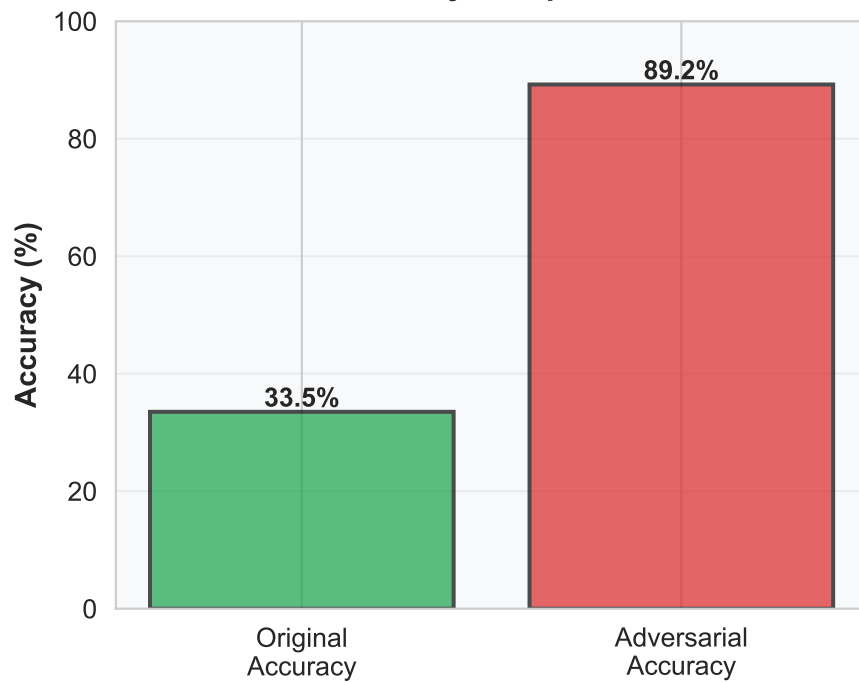
The PGD attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 33.52% to 89.23%, indicating the current robustness posture is high risk.

Assessment Narrative

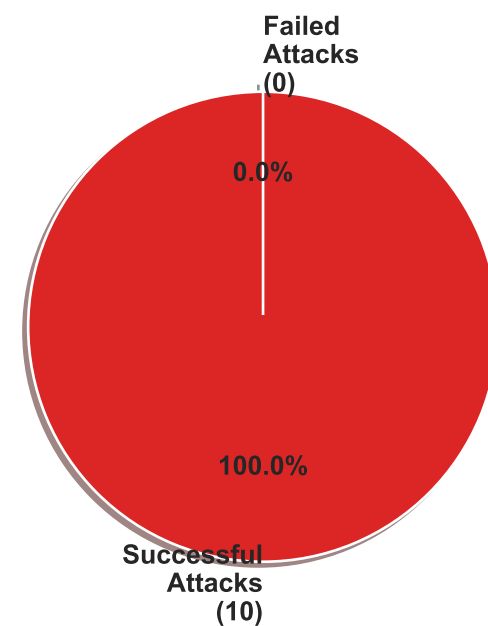
Successfully executed PGD attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 33.52%, Average adversarial accuracy: 89.23%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis

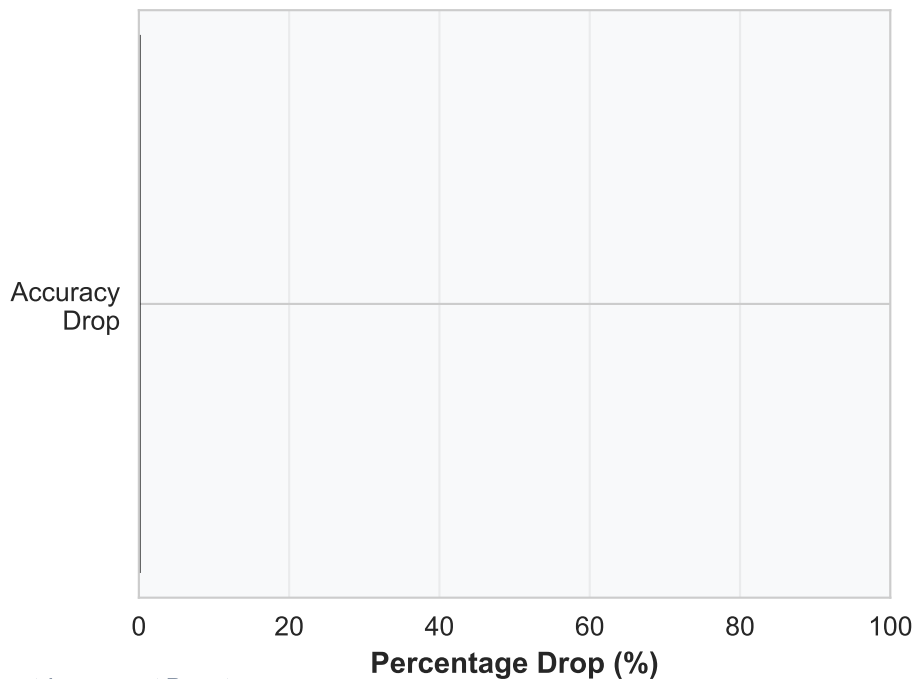
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

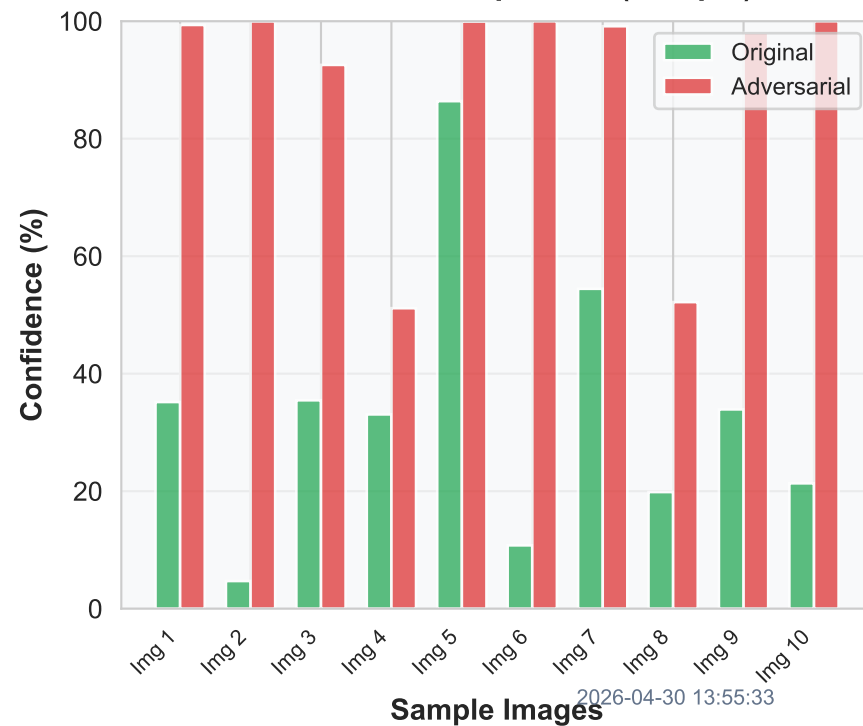
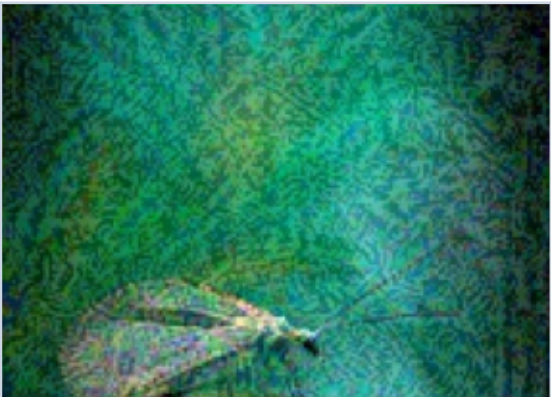


Image Evidence and Predictions

Images 1-4 of 10



Attack Successful

Original image: lacewing.JPEG
Adversarial image: lacewing.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: stole
Original confidence: 35.47%
After attack predicted as: walking stick, walkingstick, stick insect
Adversarial confidence: 92.56%



Attack Successful

Original image: ladle.JPEG
Adversarial image: ladle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: ladle
Original confidence: 86.39%
After attack predicted as: nematode, nematode worm, roundworm
Adversarial confidence: 99.94%



Attack Successful

Original image: space_shuttle.JPEG
Adversarial image: space_shuttle.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: thatch, thatched roof
Original confidence: 10.78%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 100.00%

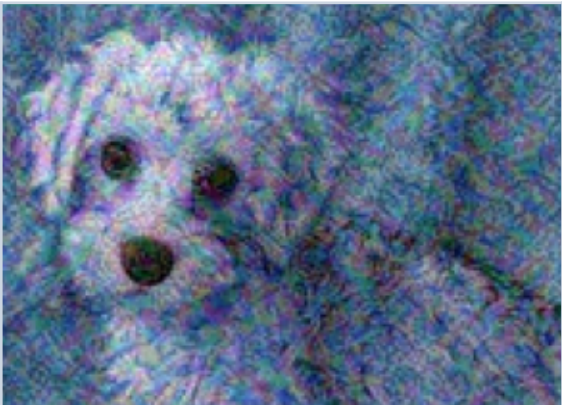


Attack Successful

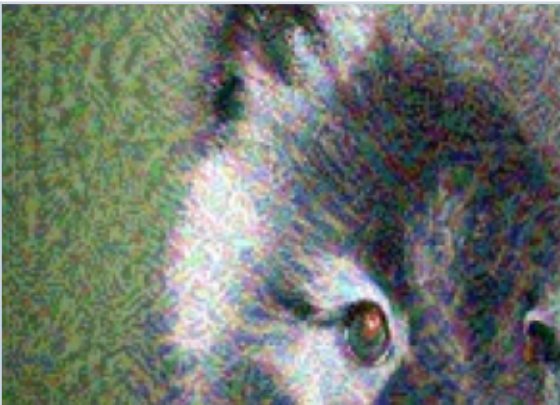
Original image: basenji.JPEG
Adversarial image: basenji.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: starfish, sea star
Original confidence: 21.34%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 100.00%

Image Evidence and Predictions

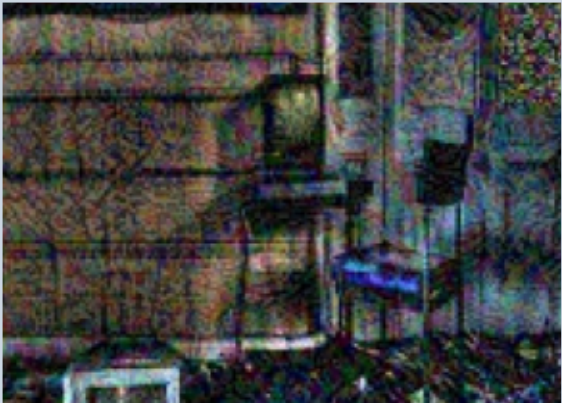
Images 5-8 of 10



Attack Successful
Original image: Dandie_Dinmont.JPEG
Adversarial image: Dandie_Dinmont.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: stole
Original confidence: 33.07%
After attack predicted as: scuba diver
Adversarial confidence: 51.15%



Attack Successful
Original image: keeshond.JPEG
Adversarial image: keeshond.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: groenendael
Original confidence: 54.45%
After attack predicted as: spider web, spider's web
Adversarial confidence: 99.14%



Attack Successful
Original image: screen.JPEG
Adversarial image: screen.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: shopping cart
Original confidence: 33.92%
After attack predicted as: spider web, spider's web
Adversarial confidence: 97.99%



Attack Successful
Original image: library.JPEG
Adversarial image: library.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: maze, labyrinth
Original confidence: 35.18%
After attack predicted as: menu
Adversarial confidence: 99.36%

Image Evidence and Predictions

Images 9-10 of 10



Attack Successful

Original image: sombrero.JPEG

Adversarial image: sombrero.JPEG (attacked)

Outcome: Attack Successful

Model predicted as: knot

Original confidence: 4.71%

After attack predicted as: nematode, nematode worm, roundworm

Adversarial confidence: 99.97%



Attack Successful

Original image: Lakeland_terrier.JPEG

Adversarial image:

Lakeland_terrier.JPEG (attacked)

Outcome: Attack Successful

Model predicted as: cardigan

Original confidence: 19.85%

After attack predicted as: tusker

Adversarial confidence: 52.19%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed PGD attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 33.52%, Average adversarial accuracy: 89.23%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques